

Research Question

How can AI enhanced personal medicine and treatment plans be beneficial to the elderly population?

Abstract

- The rising rates of **chronic illness** and **medication use** in aging populations are creating pressure on healthcare systems.
- Elderly patients often require **complex medication regimens**, which has been shown to increase their risk of **adverse drug reactions (ADRs)**.
- Standardized treatment protocols** can overlook patient-specific factors, such as a difference in how the patient metabolizes medication or how they respond to treatment
- Artificial Intelligence (AI)** and **machine learning** provide ways to analyze large datasets and find common patterns. Using health records and demographics, these technologies allow the **prediction of ADR risk for each individuals**
- There is a clear need for **personalized medicine** that adapts to everyone, especially among the elderly population.

Hypothesis

- Investigate whether machine learning models can accurately predict the risk of ADRs in elderly patients.
- Show how AI-driven prediction models can support elderly patients through personalized care through proactive treatment adjustment.

Methods

Variables Used

- Age**
Affects drug distribution, metabolism, and excretion, increasing the risk of ADRs
- Gender**
Included to observe if the model recognizes any pattern differences
- Number of Unique Diagnoses**
More diagnoses → high multimorbidity → increased risk of ADR
- Number of Unique Procedures**
Acts as a proxy for hospital care intensity
- Number of Unique Medications**
Many medications increases the risk of drug interactions, thus increasing the risk of ADR
- Subject ID**
Patient identifier

Enhancing ADR Prediction in the Elderly with Machine Learning



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Key Results

Model	Precision	Recall (accuracy)	F1	Model	Accuracy
XGBoost	0.81	0.79	0.79	XGBoost	0.79
Linear Regression	0.82	0.80	0.80	Linear Regression	0.80
Random Forest	0.82	0.83	0.83	Random Forest	0.83

- Table 1. Performance Metrics of Macro-Averages for each model (classes are weighted i.e balanced)
- Random Forest model achieved the highest performance
- Table 2. Overall Accuracies for each model

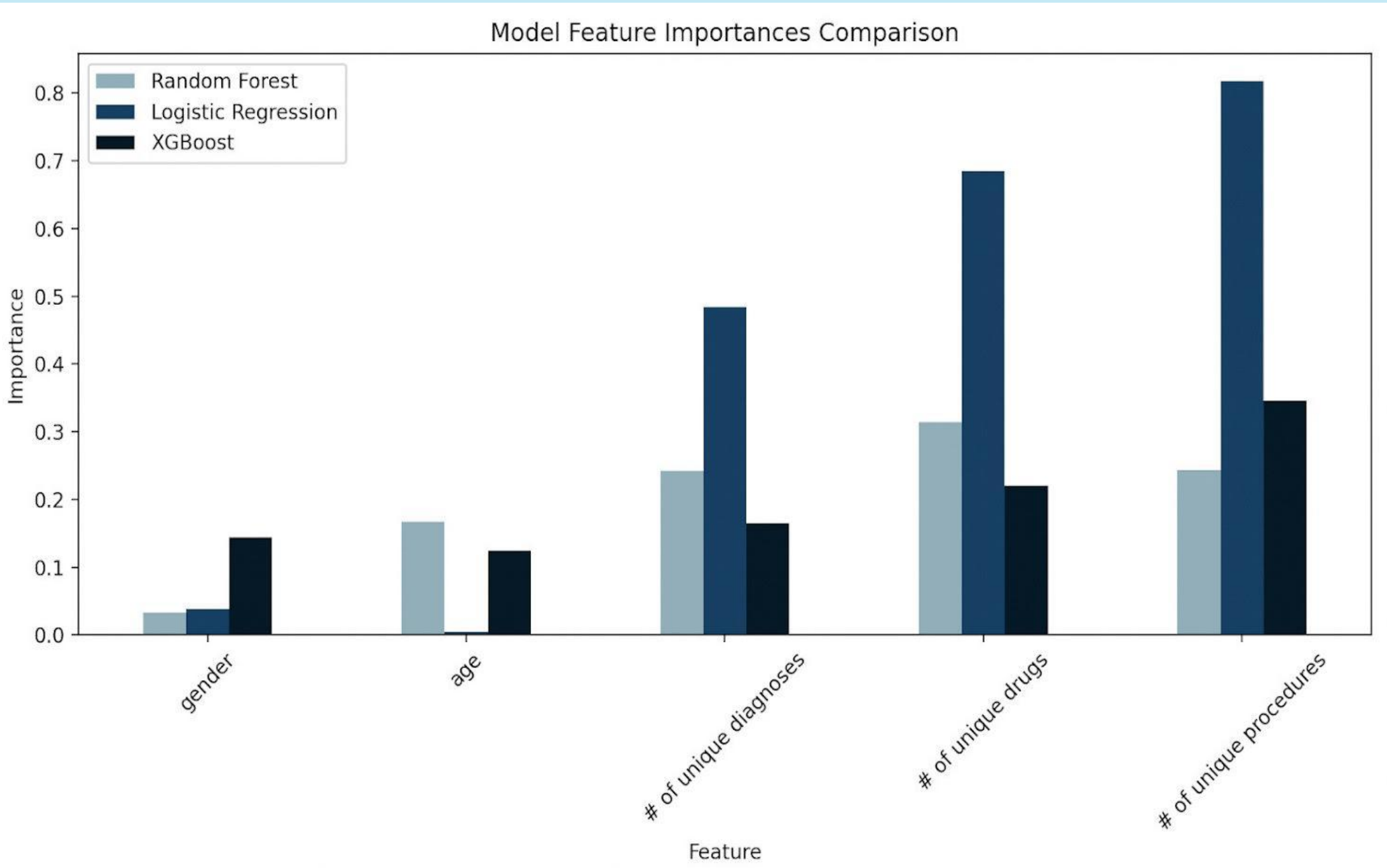
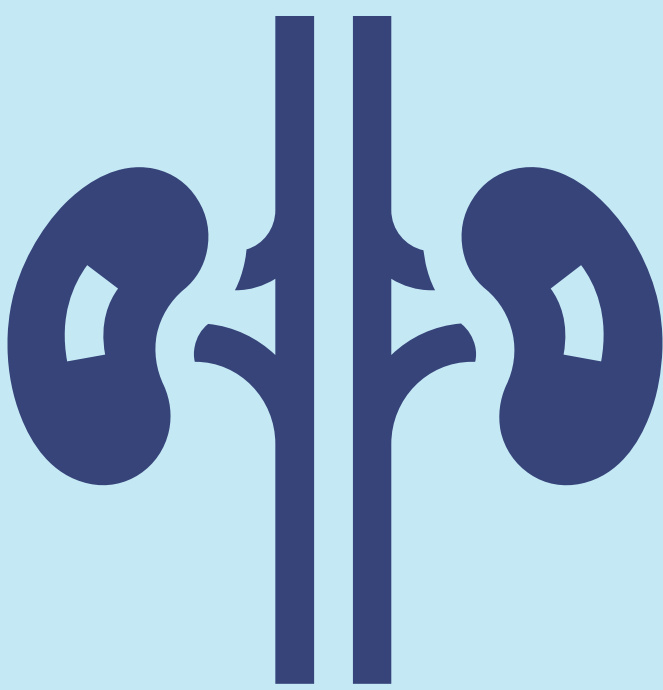
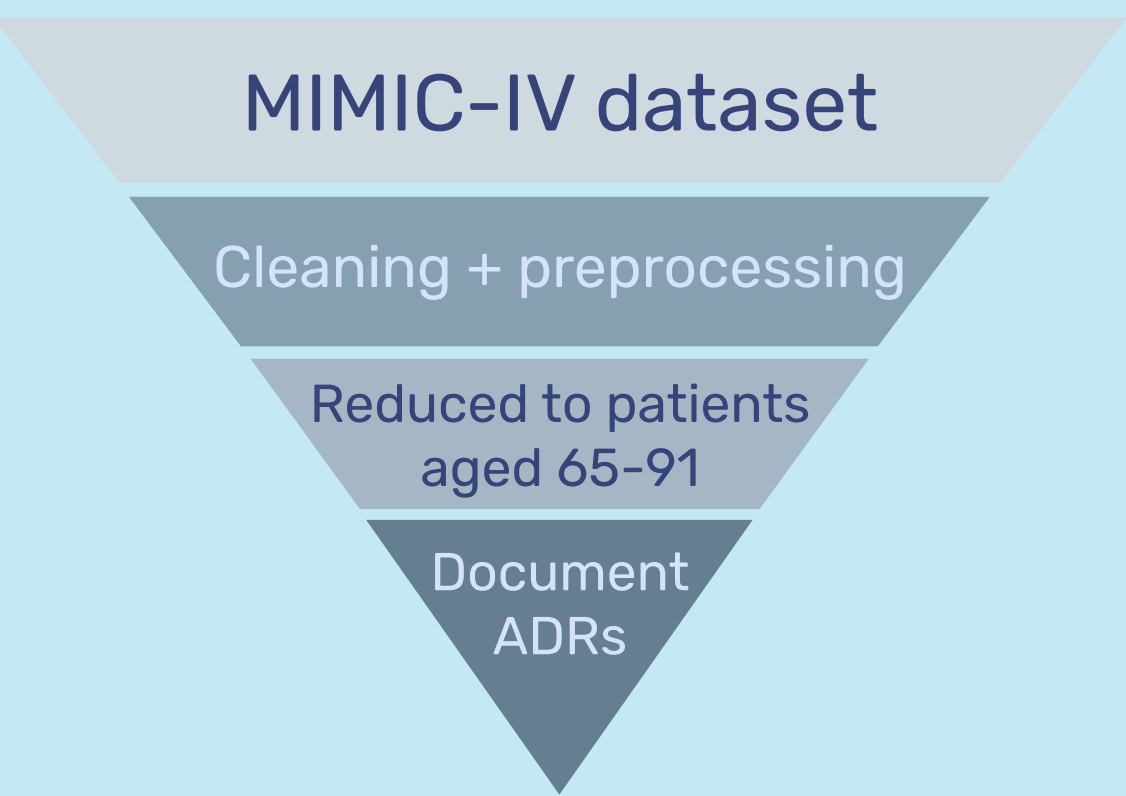


Fig. 2. Model Feature Importances Comparison chart - showing how much each property of the patients' data affects each model's result.

Interpretation / Discussion

- Machine learning models, particularly ensemble models like Random Forest exhibit strong potential in predicting adverse drug reactions (ADRs) in elderly patients
- The findings reinforce the existing clinical understanding that polypharmacy and complex, numerous medical procedures are related to an increase in ADR risk
- Age and gender are the least important, likely due to the age restrictions on the dataset (65-91 years)

Data Filtration



Models Evaluated

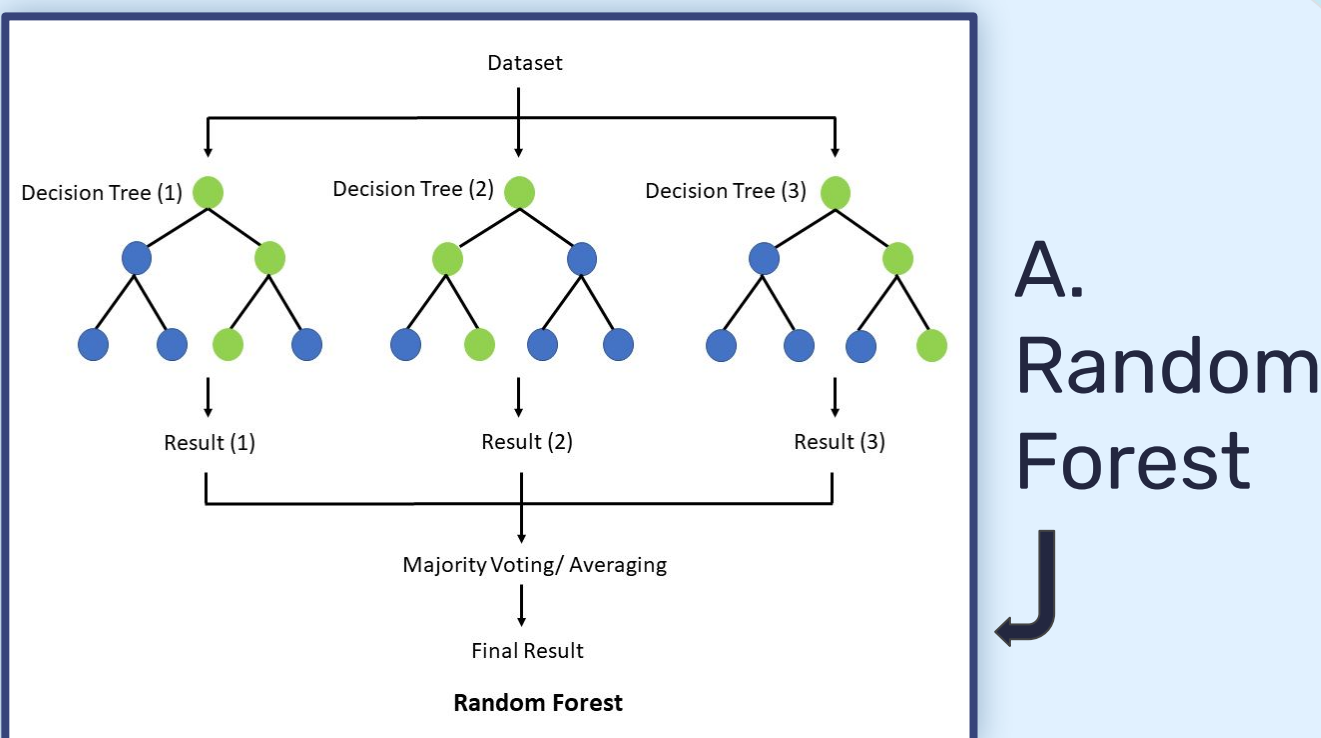


Fig. 4. Diagram showing how the Random Forest model makes predictions. Image by TseKiChun, via Wikimedia Commons, licensed under CC BY-SA 4.0.

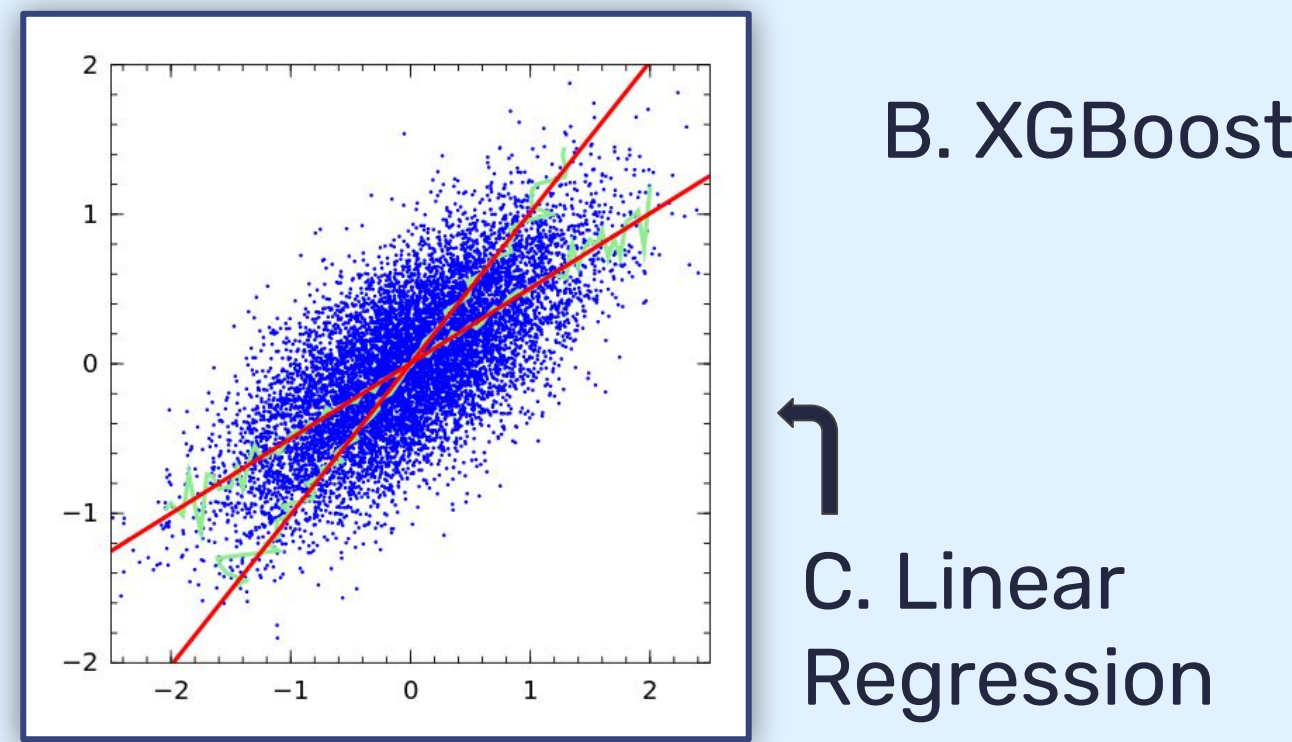


Fig. 5. General example visualization of a linear regression model. Note: image shows an "error-in-variables" regression, which is not used in this study. Image by Rickedanielson, via Wikimedia Commons, licensed under CC BY-SA 4.0.

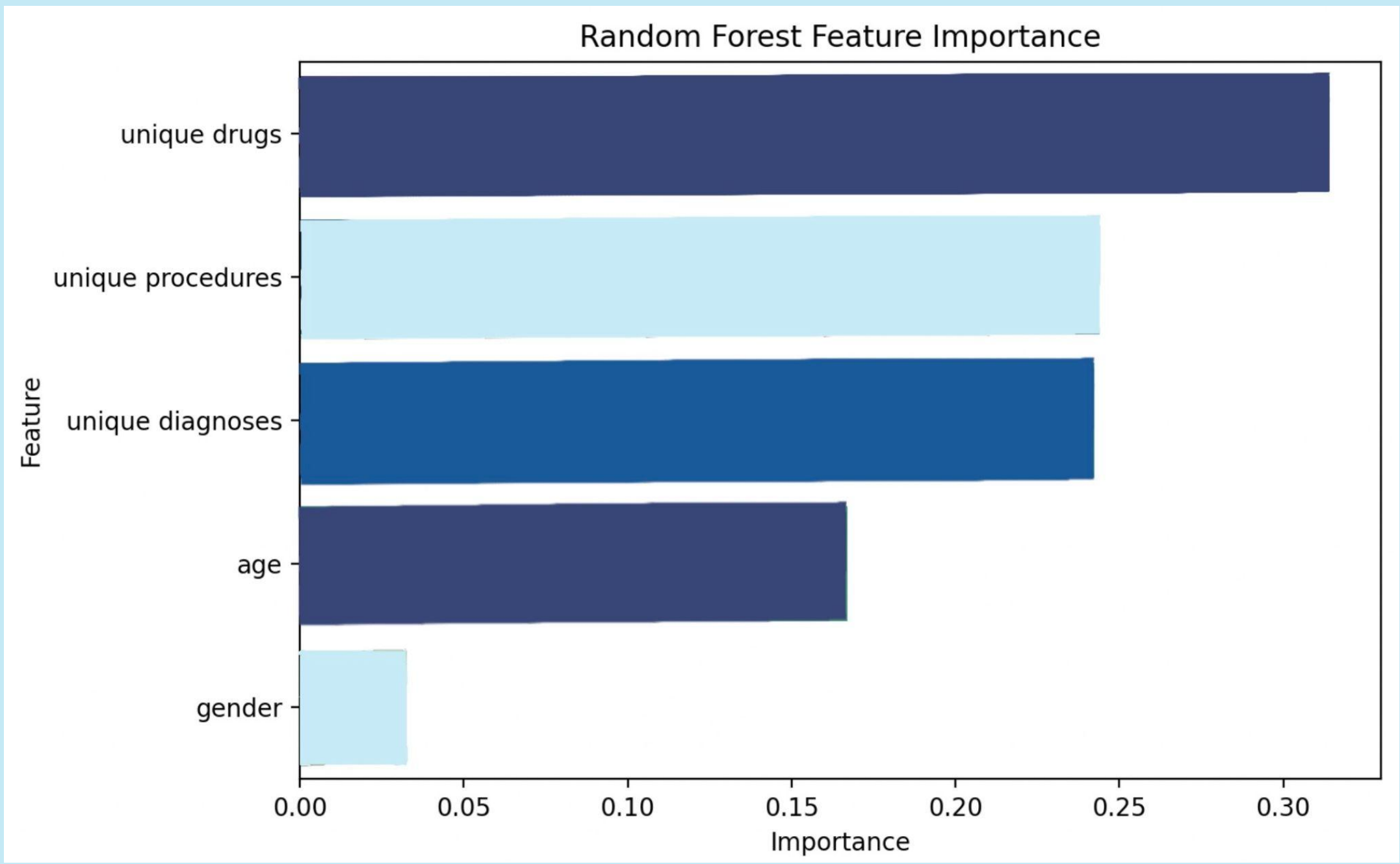
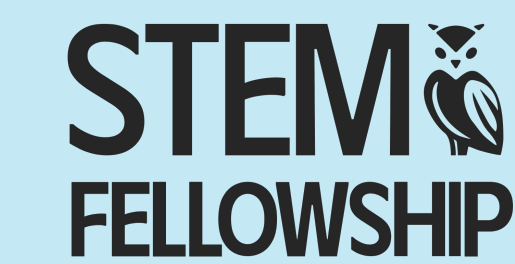


Fig. 3. Isolating the importance values for the Random Forest model

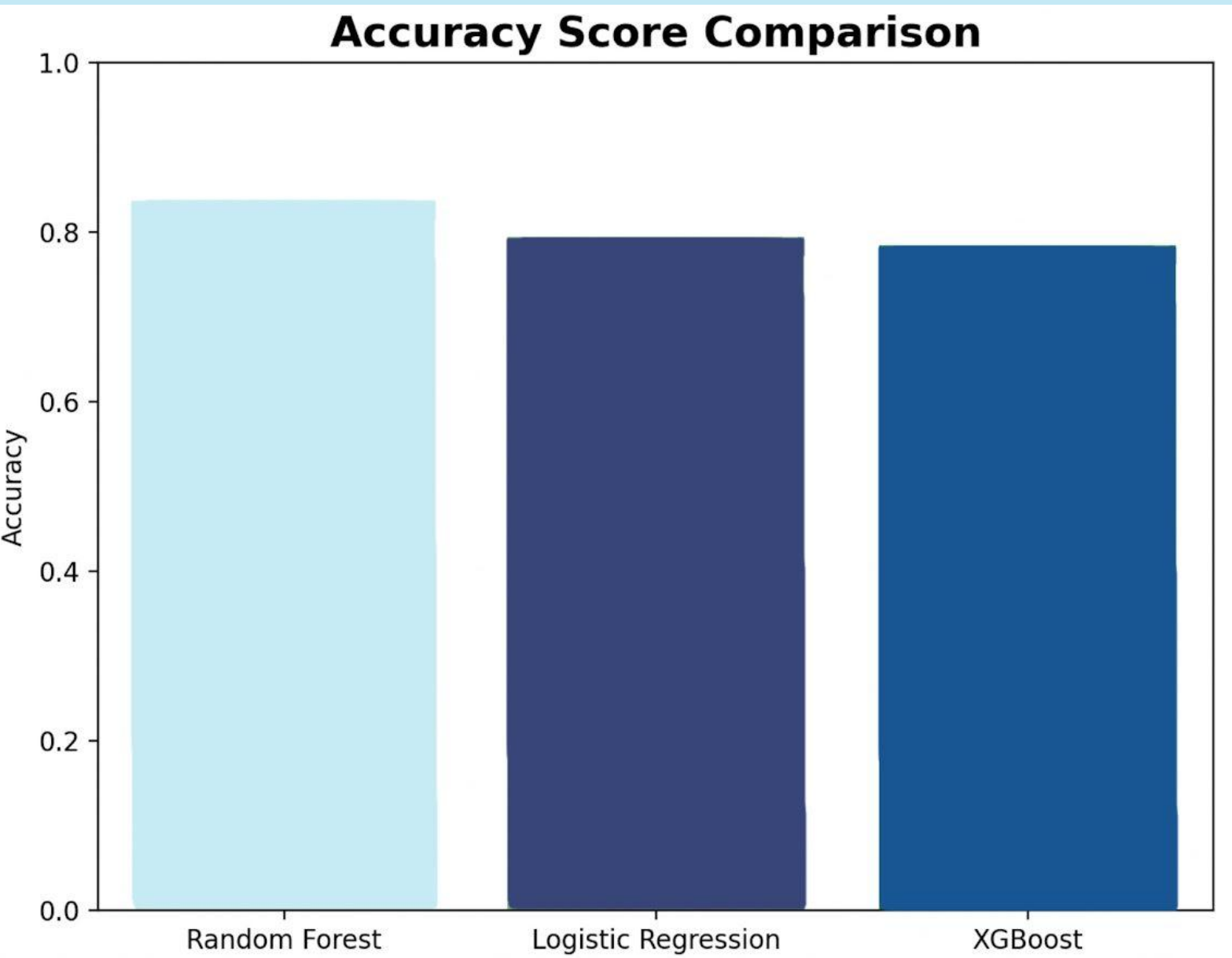


Fig. 1. Bar graph representation of Table 2. X-axis is represented by 3 different models: Random Forest (0.83 accuracy), Logistic Regression (0.80 accuracy), and XGBoost (0.79 accuracy).

Key Findings

- Our research suggests that machine learning algorithms can **accurately identify elderly patients' risk of ADR**
- The reliability of our results supports the move towards personalized care in healthcare systems, particularly geriatrics.
- The **Random Forest** outperformed the other models, consistently performing the best and displaying the **highest feature interpretability**.
- Clinicians should watch for high ADR risk factors, such as polypharmacy and the patients number of procedures

Challenges

- The **MIMIC-IV dataset** used in this research came solely from critical care settings, thus results may not generalize to **outpatient or long-term care settings**
- The machine learning models relied on **retrospective data**, so real-time risk factors or causal relationships may not have been captured
- Other important variables, including genetic predispositions, patient lifestyle, or wearable health data (e.g., Apple Watch) were not included in our research

Future Directions

- Analyzing **wearable health data** and **genomic information** may increase the accuracy of machine model predictions in the future.
- Future research would validate these models in prospective studies across a **variety of patient settings**.
- AI holds a strong potential to improve healthcare by helping clinicians quickly identify risks, personalize treatments, and ultimately **save lives**.
- The development of **interpretable and clinician-friendly AI tools** is very important to ensuring these technologies are trusted and used to their full potential.

References

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